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Article information

Article title

AI-Based Resume Screening and Analytics: Fairness Achievement and Bias Migration from Cleaning to Predictive Analysis using machine learning

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Keywords

Resume Screening; predictive Analysis; Candidate-Job Matching, Fairness Achievement and Bias Migration.

Related research article

None

For a published article:

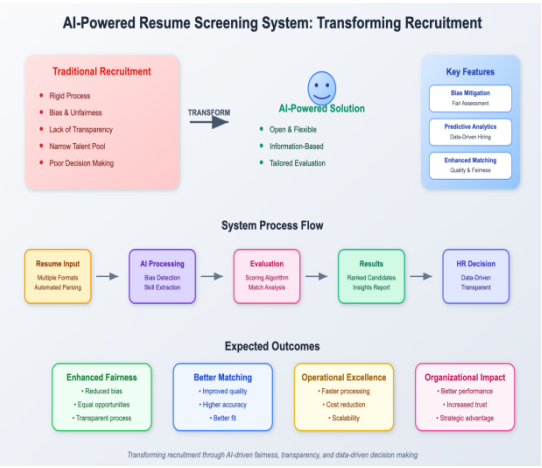
None

Abstract

Although the traditional recruitment process is vital in achieving organizational success, it tends to be rigid, unfair and biased, thereby resulting into lack of transparency and narrow choice of great talent. The paper is an attempt to overcome this gap, presenting an open, flexible, and information-based resume screening system that depends on AI. It enables HR departments to tailor their evaluation of candidates, which improves the quality and fairness of matching candidates with just predictive analytics and bias remedies available in candidates' pools, promoting better decision-making accuracy and trust. Some of the main characteristics are

- bias mitigation strategies incorporated into fair assessment
- predictive analytics to assist hiring on a data-driven basis, which leads to the enhancement of overall hiring processes and organizational performance.

Graphical abstract



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Specifications table

Subject area	Computer Science
More specific subject area	Data-Driven Recruitment and Resume Screening Systems
Name of your method	Adaptive Resume Screening and Candidate Ranking using Predictive Analytics
Name and reference of original method	None
Resource availability	None

Background

Traditional manual resume screening is a major challenge in the Human Resources departments given the current competition in jobs, not only is it time consuming, human prone to error, biased and in most instances fails to deliver on high volumes, different languages and changing demands in the industry. In order to overcome them, this project suggests a machine-learning-based solution, "AI-Based Resume Screening and Analytics: A Comprehensive Data Processing Pipeline with Data Cleaning, Exploratory and Predictive Analysis." This solution is meant to revolutionize the recruitment process and automate data extraction, profile analysis of the candidate in regard to the particular job requirements and offer a predictive score of how well a candidate fits in to the job. With the opportunity to seamlessly incorporate machine learning into the screening process, the system will provide a quicker, more informed, and accurate screening process with the help of which the HR professional can concentrate on more strategic and high-quality applicants, as well as to eliminate the problem of unconscious bias, overdependence on past findings, and inability to comprehend multilingual resumes.

Method details

Literature Review

Various prediction models are applied in AI-based resume screening and analytics, including Decision Tree, Random Forest, Naïve Bayes, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Logistic Regression (LR), Gradient Boosted Trees (GBT), XGBoost, and AutoGluon, often evaluated for tasks like candidate suitability prediction and performance assessment (Jagan Mohan Reddy et al., 2020; Luh Putu Saraswati Devia Jayanti et al., 2022; Yashomandira Kharde, et al., 2025; Samuel Akpan Robinson, J., et al 2024; Awujoola J Olalekan et al., 2021; Achmadnoer, 2017; Crane Jr, Leland, et al., 2016; Emiliano Mankolli, et al., 2023; Dana Pessach et al., 2024; pek Golbol Pekdas et al., 2024; Dr.Somanchi Hari Krishna et al., 2024; Dipti Theng et al., 2020; Ishraq Hatif Abd Almajed, 2024; G. Sudha et al., 2021; Ibrahim Alkhazraji, 2020; Sankhadip Kundu, 2024). While models like Random Forest and SVM frequently show high accuracy (e.g., Luh Putu Saraswati Devia Jayanti et al., 2022; Awujoola J Olalekan et al., 2021; Ibrahim Alkhazraji, 2020) and strong recall, especially with complex recruitment data, they can be computationally intensive or less interpretable (Luh Putu Saraswati Devia Jayanti ets al., 2022). Conversely, simpler models like Naïve Bayes are faster but may suffer from assumptions of feature independence, leading to lower performance (Jagan Mohan Reddy et al., 2020; Luh Putu Saraswati Devia Jayanti et al., 2022; Sankhadip Kundu, 2024). The integration of machine learning with techniques like genetic algorithms and NLP (e.g., BERT) aims to boost productivity, minimize bias, and enhance decision-making, with some approaches achieving very high accuracy (Yashomandira Kharde, et al., 2025; Emiliano Mankolli, et al., 2023). However, common challenges across these methods include a strong reliance on high-quality and sufficient data, potential for overfitting, implementation complexities, and the ongoing trade-off between model interpretability and predictive accuracy, alongside ethical considerations like privacy and fairness

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(Yashomandira Kharde, et al., 2025; Samuel Akpan Robinson, J., et al 2024; Awujoola J Olalekan et al., 2021; Dana Pessach et al., 2024; Dipti Theng et al., 2020; Ishraq Hatif Abd Almajed, 2024; G. Sudha et al., 2021; Ibrahim Alkhazraji, 2020; Dr.Somanchi Hari Krishna et al., 2024).

Efforts in AI-based recruitment increasingly focus on Explainable AI (XAI) and bias mitigation to enhance transparency, fairness, and trust in hiring processes (Ravi Kiran Magham, 2024; Dr.D.Anitha Kumari et al., 2024; Köchling et al., 2023; Christopher G. Harris, et al., 2023; Django Beatty et al., 2024; Alejandro Peña et al., 2023; Alejandro Pe˜na, 2020). Techniques like SHAP with interpretable models (Decision Tree, Random Forest) provide insights, while methods such as anonymization of CVs, sensitive attribute removal via algorithms like SensitiveNets, and bias correction tools (e.g., IBM AI Fairness 360, Microsoft Fairlearn) are employed to address gender, age, and other forms of discrimination (Ravi Kiran Magham, 2024; Christopher G. Harris, et al., 2023; Django Beatty et al., 2024; Alejandro Peña et al., 2023; Alejandro Pe˜na, 2020). While these approaches promise objectivity, efficiency, and improved diversity, challenges remain, including potential trade-offs between accuracy and interpretability, the complexity and cost of developing and implementing robust XAI and bias mitigation systems, reliance on high-quality training data, the emergence of new or subtle biases, and difficulties in accounting for soft skills or integrating solutions into existing large-scale systems (Ravi Kiran Magham, 2024; Rishi Jain et al., 2023; Dr.D.Anitha Kumari et al., 2024; Köchling et al., 2023; Christopher G. Harris, et al., 2023; Django Beatty et al., 2024; Alejandro Peña et al., 2023; Alejandro Pe˜na, 2020).

The application of Explainable Artificial Intelligence (XAI) and Natural Language Processing (NLP) is increasingly vital in recruitment to enhance transparency and mitigate bias. Methodologies integrate models like Tf-Idf with BigBird to balance computational efficiency with contextual understanding for documents such as resumes, although transformer models like BigBird demand significant computational resources (Prasanna Parasurama, 2023). Bias mitigation techniques, including post-processing (e.g., Decouple Scale), pre-processing (e.g., Groupwise z-Normalization), and dictionary-based approaches, aim to improve fairness through scalable and easily implementable methods, despite potential retraining needs or contextual misunderstandings (ALESSANDRO FABRIS et al., 2024). Other schemes combine LSA, BERT, and SVM for enhanced prediction precision, explainability, and stability, though BERT's complexity and data demands can be limiting (Xiaoguang Tian et al., 2022). Furthermore, Transparent Machine Learning (TML) methodologies focus on source code representation and operational semantics for human interpretability and AI safety, albeit with high computational requirements and implementation challenges (Dustin Juliano, 2019). The design of transparent recruitment modules and the use of techniques like LIME and SHAP, coupled with stakeholder involvement, also contribute to increased accountability and bias recognition, despite facing issues of complexity and potential misinterpretation of results (DORNADULA, 2016).

Proposed Algorithm

1. Bias migration and fairness achieving, Data preprocessing stage

The bias mitigation and fairness method to resume screening entails a tedious preprocessing of data whereby sensitive personal identifiers such as gender, race, age, and school affiliations are strictly removed leaving only merit- and job-based qualities such as the experience, skill, and certificates to form an impartial dataset. Afterward, fairness should be evaluated over time, through computing the True Positive Rate (TPR) of each group and comparing, graphically depicting, or similar group to detect a disparity in the rates, and through the model adjustment via feature engineering, re-weighting, or adversarial debiasing by re-comparing, repeating as long as the model is not validated as fair enough to provide equal opportunities.

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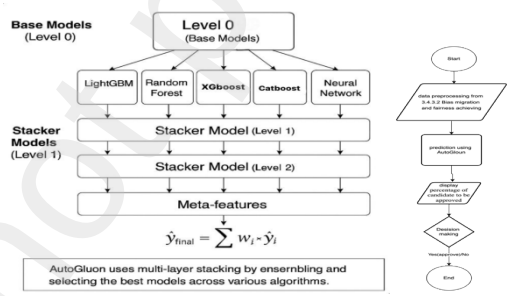
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$$\text{TPR} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$



2. Prediction model using AutoGloun model

The predictive analysis system based on AI is deployed in the framework of Amazon's AutoGluon, which is selected because of its built-in machine learning and an excellent ensemble schema. Data preparation and preprocessing are the first stage of work where raw data of candidates is carefully transformed, scaling of numerical data to form numerical data, conversion of categorical data, and missing values are handled to work with data as input to models. AutoGluon is then used to automatically select, optimize and combine a wide variety of machine learning algorithms, such as XGBoost, LightGBM, CatBoost, and neural networks, in a deep stacking structure to produce accuracy probability estimates of the best predictions by each of the models. After the prediction, these scores are shown to the HR, explaining the predictions and making each of them transparent. This will help the HR professionals make a data-driven hiring decision, that is, approve the qualified candidates to be considered and all results addressed to be able to allow effective recruitment process in the future by conducting the audit.



PROTOTYPE IMPLEMENTATION

1. Data Preprocessing of "resume_data.csv" Dataset

Preprocessing of the given dataset, such as the resume_data.csv, representing the profile of candidates and job requirements, was performed by a detailed preprocessing pipeline to make it fit a prediction task. The first cleaning was done by lowercasing and the elimination of special characters within column fields. Important data such as the education and level of experience were captured in the job requirement fields and the educational and work histories of the candidates were checked to obtain the most recent degree and the years of experience that a candidate has which was in turn matched with the job requirement fields to obtain the experience adequacy percentage. To augment the set of features with semantic knowledge, the pre-trained Sentence-BERT model was used to compute the semantic similarity scores, which evaluated the adequacy of the candidate skills, positions, and education in comparing with the job advertisements. And columns of the textual type, such as abilities and degrees, used Sentence-BERT representations and TF-IDF vectors. The Gaussian values of the GPA obtained out of raw education outcomes were recoded into a 5-point scale of categorization. Considering that the TF-IDF features were extremely high-dimensional, Principal Component Analysis (PCA) technique was used to simplify the features to a lower dimension in case that most likely covers 95 percent of the variation in data. The overall list of features that followed this combination, the numerical data, semantic similarities and the reduced with PCA text data, was then divided into the training and test sets to provide a good basis of building up job-resume matching predictive models.

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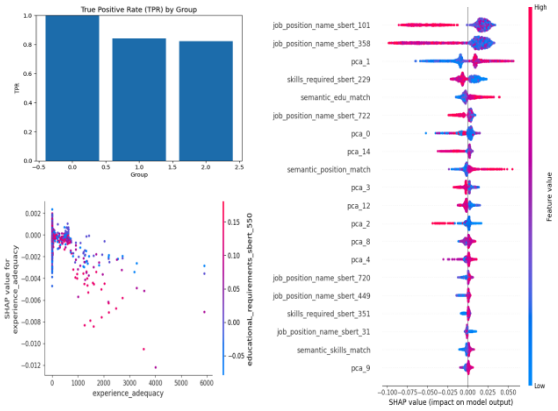
2. Prediction Model: AutoGlaun

This paper used the TabularPredictor capability of AutoGluon to build a good predictive model of the matched_score value, a numerical score used to match the candidate resume to a job description that can be conceptualized as a regression task, the objective of which is to minimize the R-squared (r2) value. The model applied the best_quality preset so that it can be widely trained and feature engineering, tuning hyperparameters, and innovative ensemble methods were possible. The system successfully handled heterogeneous data, consisting of structured fields (degree or education, experience), as well as unstructured textual information (skills, career interests), and all the sets of feature extractors in the system were enabled, thanks to AutoMLPipelineFeatureGenerator. Some of the state-of-art algorithms, which were comprised in the model ensemble, were the Gradient Boosting Machines (GBM), the CatBoost (CAT), the XGBoost (XGB), and the neural networks (NN_TORCH), Random Forests (RF), K-Nearest Neighbors (KNN), FastAI deep learners, with their hyperparameters optimized per each algorithm separately. Additional advanced forms of ensembling also enhanced the model, using 10-fold bagging to reduce variance, stacking at two levels to absorb previous model predictions, as well as dynamic stacking or dynamic_stacking=True to sensibly select the most promising models, and to do so using automatic hyperparameter tuning.

Method validation

1. Bias migration and fairness achieving

In assessing the fairness and reducing biasness, the screening model went the dual approach which is to say, in addition to a strict first round preprocessing, it had an after-the-fact (post-hoc) fairness checking. The preprocessing stage was done promptly to clean up the personally identifiable and bias-generating characteristics (e.g., gender, age, institution names) and only the merit and role-related characteristics (e.g., job experience, skills, qualifications, certifications, and semantic similarity scores between resumes and job descriptions) were retained. The interpretable features, such as semantic skill match and experience adequacy, were included in the modeling when the text was transformed into SBERT embeddings and PCA approach was used to support interpretability and avoid overfitting. Based on that, fairness was assessed on a test set based on True Positive Rate (TPR) analysis across subgroups and in all groups; it demonstrated high and near values of the TPR (e.g. 0.82-1.0) which averts violating the Equal Opportunity principles. The effect of experience_adequacy on predictions was further assessed to be smooth and not dependent on arbitrary cut-offs and most importantly, in SHAP (Shapley Additive explanations) analysis, position and skills SBERT embeddings and semantic matching scores were among the most powerful features. This established that the model choices were made depending on meeting the qualifications and not demographic attributes, and the sensitive characteristics were essentially removed as part of the initial preprocessing, making it out of the top predictors. Overall, the results indicate the effectiveness of the model to produce fair, generalizable and transparent results using only job-relevant data and, therefore, its potential to conduct unbiased automated screening in practical use.

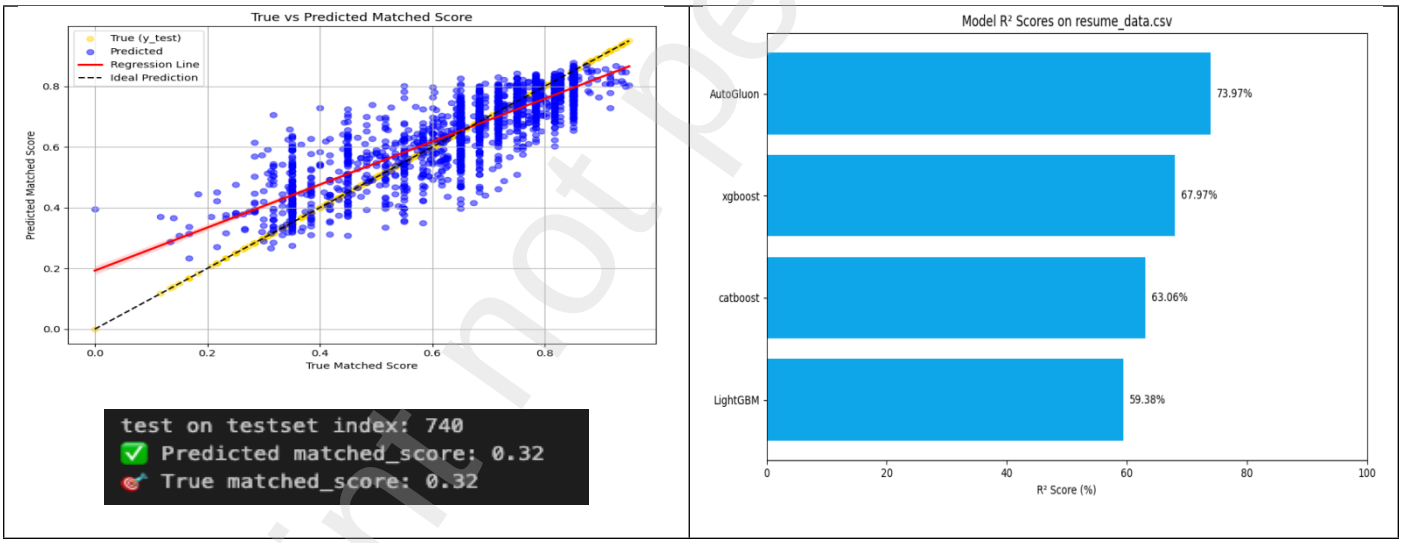


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2. AutoGluon model

The following experiment compared four machine learning techniques, AutoGluon, XGBoost, CatBoost, and LightGBM, and trained them on the dataset of job-resume matches hereby named resume_data.csv to rank jobs and resumes using R2 score as the principal regression metric. AutoGluon was fine-tuned drastically on this regression task (problem_type='regression', eval_metric='r2') by setting TabularPredictor feature_generator to consist of a lengthy feature generator to accept both raw text, n-grams, numeric, datetime, and categorical values. It was also able to use custom hyperparameters on different model families (GBM, CAT, XGB, NN_TORCH, RF, FASTAI, KNN) and use even more complex ensembles (10-fold bagging, two-level stacking, dynamic stacking dynamic_stacking=True) and automatic hyperparameter tuning (hyperparameter_tune_kwargs='auto'). The other models had default settings, or the recommended settings. The outcomes indicated that AutoGluon performed much better than the others with R2 score of 73.97% whereas the XGBoost performed second and was 67.97%, and the CatBoost and LightGBM ranked third with 63.06 % each. Such high results of AutoGluon are explained by the strong AutoML framework, extensive fine-tuning, and sophistication in dealing with features of quite different types as well as the significant ensemble methods that altogether maximize predictive power and generalization. XGBoost was also performing comparatively well, since it was both efficient and could regularize, and CatBoost, in turn, displayed satisfying outcomes because of its capabilities of handling categorical features, and, in spite of its overall efficiency, LightGBM did not show the best R2 in this study.



Limitations

Although it achieved its targets, the project still faced some of the following limitations, e.g., the absence of computing resources necessitating optimization and experimentation of the resource-processor AutoGluon framework. What is more, the poor quality and abundance of the dataset with a lack of industry, job type, and linguistic diversity, variable organization of resumes and availability of details limited the capacity of the model to present generalizable patterns and to be effective on unfamiliar real-life formats. Although there was an attempt to offset the bias via interpretability, due to the nature of bias as a complex issue in machine learning models, along with non-live deployment of the system, the effects of fairness metrics could not be clearly evaluated. Also, the fact that the system made use of the structured data and therefore, utilized the classic approach to extracting keywords restricted its ability to grasp a natural language: it would miss out on important contexts, such as the ability to perform under leadership or the cultural compatibility.

Ethics statements

The report does not involve human participants or data from social media platforms.

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CRediT author statement

Chang Kai Lok: Research, Conceptualization, Metghodology, implementation, Writing – Original Draft, Writing – Review & Editing

Dr. Ng Kok Why: Writing – Review & Editing, supervision

Acknowledgments

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Supplementary material *and/or* additional information [OPTIONAL]

None

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